

Chapter 62. Diaphragmatic Pacing

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Diaphragmatic Pacing: Introduction

Chronic respiratory failure requiring ventilatory support is usually caused by severe derangements in the function of the lungs, chest wall, and/or respiratory muscles. Mechanical ventilation is the only feasible option for most patients with these disorders. In certain patients, however, the function of the respiratory apparatus is completely intact except for lack of adequate nervous output from the respiratory centers in the medulla (central hypoventilation syndrome [CHS])¹⁻³ or interruption of electrical signals from the medulla to the respiratory motoneurons in the spinal cord, which innervate the major inspiratory muscles (cervical spinal cord injury). These patients can be offered an alternative means of respiratory support by diaphragmatic pacing (DP), a more natural and physiologic form of breathing.³⁻¹²

Although straightforward in concept, DP required several significant scientific developments before becoming a clinically useful modality. A brief historical perspective provides some insight into the evolution of DP and understanding of its clinical utility.

It was more than two centuries ago that Caldani¹³ first observed that diaphragm movement could be achieved by electrical stimulation of the phrenic nerve in animals. In the first human demonstration (1818), Ure¹⁴ stimulated the phrenic nerve and restored breathing in a criminal immediately following execution. In the latter portion of the nineteenth century, the application of moistened sponges over the outer borders of the sternocleidomastoid muscles resulted in activation of the phrenic nerves, and this technique became an accepted method of restoring ventilation.^{15,16} Duchenne found phrenic nerve stimulation to be the best method of producing natural respiration¹⁷ and is credited to have placed electrophrenic stimulation on a solid physiologic foundation.¹⁸ In the 1940s, Sarnoff et al^{19,20} demonstrated that ventilation could be maintained in an acute setting in patients with bulbar poliomyelitis utilizing percutaneous electrodes. The lack of implantable devices that were safe and reliable, however, limited the long-term usefulness of DP to support ventilation. Consequently, the technique of phrenic nerve stimulation was supplanted by the development of the more reliable mechanical ventilators.

In the 1960s, the critically important work of Glenn et al resolved major technological issues, which led to the development and implementation of modern-day pacing systems.²¹⁻²³ These investigators defined the appropriate patient selection criteria, preoperative assessment methodology, surgical methods including optimal electrode placement, and appropriate stimulation parameters necessary to achieve full-time ventilatory support.^{8-12,23,24} Further refinements to this early design were made by other investigators with regard to improved electrode design^{25,26} and less-invasive methods of electrode placement.²⁷⁻³⁰ As a consequence, DP has evolved into a safe and practical method of providing ventilatory support in select patient groups.

Rationale for Diaphragmatic Pacing

For patients requiring chronic ventilatory support, DP provides several advantages compared to mechanical ventilation (Table 62-1).^{5,6,11,31-35} Although the realized benefits of DP are subjective and vary between individuals, most patients describe an improved sense of well-being and overall health. This impression can be attributable to one or more factors. Because patients are utilizing their own breathing muscles, patients relate the sensation of more normal breathing. Negative pressure ventilation also has the potential to reduce the incidence of barotrauma and may have beneficial cardiovascular effects.³⁶ Because ventilator tubing is unnecessary, tension on the tracheostomy tube by attached tubing is eliminated, improving patient comfort.

Table 62-1: Potential Benefits of Diaphragmatic Pacing

A. Improved quality of life
1. Subjective sense of more normal breathing
a. Utilization of breathing muscles
b. Negative-pressure respiratory support
2. Improved comfort level
a. Elimination of pull of ventilator tubing
b. Negative-pressure breathing
3. Improved speech
4. Restoration of olfactory sensation
5. Increased mobility
a. Easier transport outside the home
b. Easier transfer to and from bed
6. Reduced anxiety and embarrassment
a. Elimination of ventilator noise
b. Elimination of fear of ventilator disconnection
c. Elimination of ventilator tubing
d. Daytime closure of tracheostomy
B. Reduced overall costs
1. Reduction or elimination of ventilator supplies
2. Reduced level of nursing and respiratory therapy services

Source: Used, with permission, from Dimarco.³¹

Although the volume of speech may be less with DP than with mechanical ventilation, quality of speech is often improved because ventilator interference is eliminated. Speech can also be enhanced by use of an abdominal binder.³⁷ Olfactory sensation is also restored with DP contributing to improvement in life quality.³⁸ There is some evidence that the incidence of respiratory tract infections is less with DP compared to mechanical ventilation. In a prospective clinical trial of sixty-four patients over a 20-year period, there were significantly fewer respiratory infections in DP patients compared to those on mechanical ventilator.³⁹ The patients on mechanical ventilation, however, were significantly older compared to those on DP (mean: 53 vs. 29 years old).³⁹

A life-support system that requires attachment to a mechanical ventilator by connecting tubing is extremely restrictive. Patient mobility and patient transport therefore is usually enhanced by DP. Simple daily maneuvers such as transfer from bed to chair are less cumbersome and eventful. Transport of patients outside the home is also easier. Improved mobility may allow patients to acquire gainful employment, become eligible for participation in specific rehabilitation programs, and participate in more social events.^{31,34,40–42} In a recent study, improved mobility and sense of freedom were described as the main benefits of DP.³⁸

Other concerns for many patients include social embarrassment because of ventilator noise, attached tubing, and being tethered to a machine. Many patients cannot support themselves off the ventilator and live in constant fear of disconnection. By comparison, DP is virtually indistinguishable from normal breathing by observers, and the fear of ventilator disconnection is eliminated.

The institution of DP is very expensive. With direct phrenic nerve stimulation systems, total costs can easily exceed \$100,000. Although substantially less, the total costs of the intramuscular DP system can be expected to exceed \$40,000. Once in place, however, servicing and maintenance costs are minimal.^{33,43} Despite the high initial cost, it can be argued that DP is cost effective. Many patients are transferred to less intensive and less expensive care settings.^{31,32,44} The high cost of maintenance supplies necessary with mechanical ventilators is reduced. Because most patients who undergo DP are young and require respiratory support for many years, these cost savings can be substantial.^{44–46}

Equipment

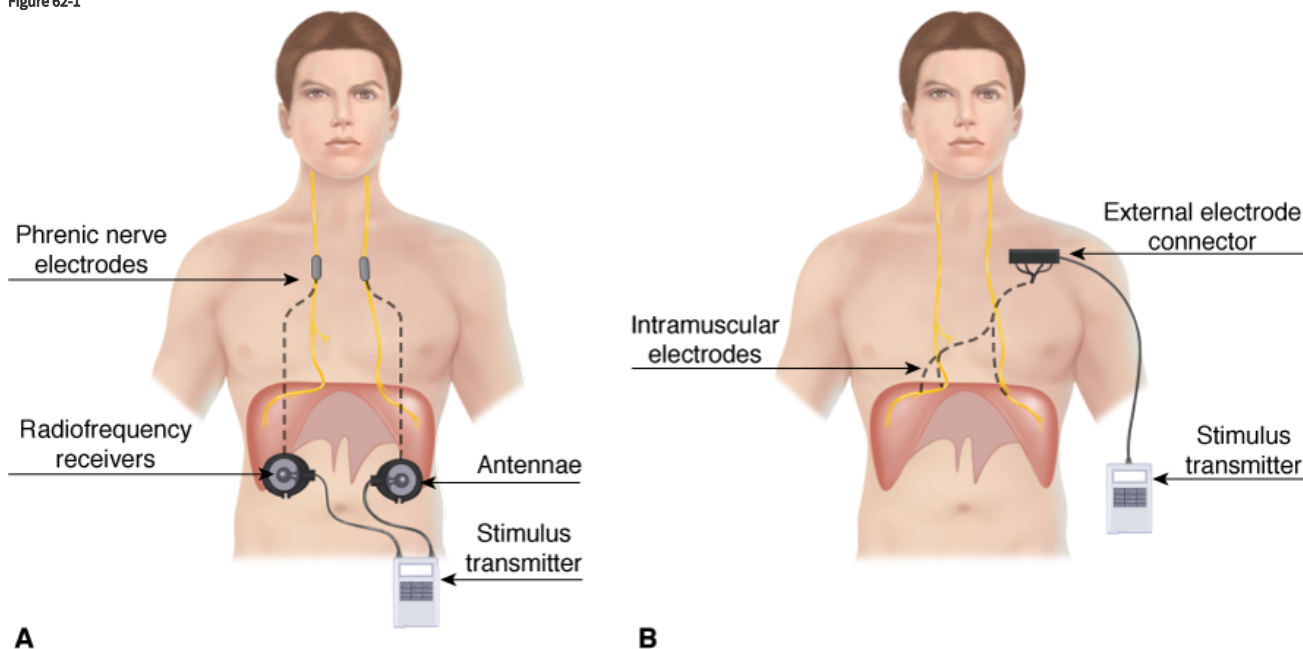
There are two basic types of commercially available DP pacing systems. In the first, electrodes are implanted directly on the phrenic nerves (direct phrenic nerve stimulation system). In another, more recently developed system, electrodes are implanted within the diaphragm (intramuscular DP

system).

Direct Phrenic Nerve Stimulation Systems

Each of these systems is very similar in design, consisting of both implanted materials and external components (Fig. 62-1A). The electrodes and radiofrequency receivers comprise the surgically implanted components. An external power supply, radiofrequency transmitter, and antenna wires comprise components outside of the body.

Figure 62-1



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Basic design of commercially available diaphragmatic pacing systems. Direct phrenic nerve stimulation system (A) and intramuscular diaphragmatic pacing system (B). With the direct phrenic nerve stimulation system, the internal components consist of a single electrode implanted on each phrenic nerve in the thorax with a wire connected to each implanted radiofrequency receiver. The external components consist of a stimulus transmitter and attached rubberized antennae, which must be positioned over the radiofrequency receivers. The receiver converts radiofrequency signals from the transmitter into electrical signals, which stimulate the phrenic nerves to activate the diaphragm. With the intramuscular diaphragmatic pacing system, two wires are implanted into the body of each hemidiaphragm near the phrenic nerve motor points. These wires are tunneled subcutaneously to a site over the chest wall where they exit the skin and are attached to an external electrode connector.

With each system, a single stimulating electrode is surgically positioned on each phrenic nerve. With unipolar systems, an indifferent electrode is also implanted subcutaneously.¹⁰ Wires are tunneled subcutaneously to connect each electrode to a radiofrequency receiver that is positioned over the anterior chest wall, usually over the lower rib cage.¹⁰ The external battery-powered transmitter generates radiofrequency signals, which are inductively coupled to the receivers via circular rubberized antennas. Each antenna must be positioned directly over each receiver and secured in place to ensure proper transmission of the signal. The transmitter signal is demodulated by the receivers, converting the radiofrequency signals to electrical signals, which are transmitted to the electrodes to stimulate the phrenic nerves and activate the diaphragm.^{10,20}

The transmitter allows adjustment of stimulus amplitude (milliamperes [mA]) and stimulus frequency (Hertz [Hz]) to modulate the magnitude of tidal volume. Respiratory rate can be adjusted by altering the train rate. Inspiratory time and inspiratory flow rate can be adjusted, in tandem, by alterations of stimulus on-time. Sigh breaths can also be provided.

Surgical techniques have been developed for both cervical and thoracic placement of phrenic nerve electrodes.⁸⁻¹² Although less invasive from a surgical standpoint, the cervical approach has significant disadvantages. First, an accessory branch from a lower segment of the cervical cord may join the main trunk of the phrenic nerve low in the neck or in the upper thorax.⁹ Activation of the cervical portion of the phrenic nerve may therefore lead to incomplete activation of the phrenic nerve and reduce the chance of successful pacing. Second, other nerves in close proximity to the phrenic nerve may also be activated, resulting in pain and/or unwanted movement.⁴¹ Finally, neck movement may displace the electrode, resulting in incomplete diaphragm activation and/or place significant mechanical stress on the nerve increasing the risk of injury. Consequently, the thoracic approach is the preferred method of implantation.^{26,32,48}

Phrenic nerve electrodes have been successfully placed thoroscopically.^{30,48} This procedure involves placement of trocars in several intercostal spaces and is technically quite demanding. Successful pacing was achieved for 12 to 14 hours/day while awake in a small group of children, primarily for management of congenital CHS. Moreover, although less invasive than thoracotomy, these patients also developed complications of pneumonia, atelectasis, and pneumothorax postoperatively. Additional studies are necessary to determine the long-term success of this procedure and its applicability to patients with cervical spinal cord injury.

Of the three commercially available DP systems, only the Avery system is available worldwide. Table 62-2 lists the technical characteristics of each device.

Table 62-2: Technical Features of Diaphragmatic Pacing Systems

Manufacturer	Avery Laboratories	Atrotech OY	Medimplant
Transmitter (stimulus generator)	Mark IV ^a	PX 244	Medimplant 8-channel stimulator
Size (mm)	146 × 140 × 25	185 × 88 × 28	170 × 130 × 51
Transmitter/battery weight (kg)	0.54	0.45 + 0.6 (12V) 0.45 + 0.046 (9V)	1.42
Rate (breaths/min)	6 to 24	8 to 35	5 to 60
Amplitude (mA)	0 to 10	0 to 6	0 to 4
Pulse width (μs)	150	200	100 to 1000
Battery life (hours)	400	160–320 (12V) 8 (9V)	24
Sigh possible	Yes	Yes	Yes
Antenna	902A	TC 27-250/80	RF transmission coil
Receiver	Model I-110A	RX 44-27-2	Implantable receiver
Size (mm)	30 (diam) × 8	49 (diam) × 8.5	56 × 53 × 14
Electrodes	Monopolar, bipolar	Quadripolar	Quadripolar
No. of receivers to stimulate both hemidiaphragm	2	2	1

^aThe Mark IV supersedes all earlier versions manufactured by Avery Laboratories. Most patients implanted with earlier models have been upgraded to the Mark IV without surgery.

Source: Used, with permission, from DiMarco.³¹

Avery Laboratories Mark IV

Glenn et al performed the basic science and clinical studies that led to the commercial application of the Avery (Commack, NY) system in the 1960s.^{49,50} The electrode consists of a semicircular platinum-iridium ribbon embedded in molded silicone rubber. The electrode contains a shallow trough for placement of the phrenic nerve. Monopolar and bipolar electrodes (recommended for patients with cardiac pacemakers) are available.⁹ The recent Avery Mark IV transmitter is lighter and provides a wider range of stimulus amplitudes than previous models.³³ An optional interface also allows biofeedback control from pulse oximetry and CO₂ monitoring.³³ Transtelephonic monitoring is available, which allows the electronic output and neurophysiologic response of the pacing system to be monitored by telephone.⁵¹

Atrotech Oy

The unique feature of the Atrotech (Tampere, Finland) system lies in its electrode technology, which consists of a four-pole electrode design.^{25,52} The electrode consists of two identical strips of Teflon fabric with two platinum buttons mounted onto each strip. Theoretically, when appropriately placed, the phrenic nerve is divided equally into four stimulation compartments. Each quadrant of the nerve, which supplies a specific set of diaphragm motor units, is stimulated sequentially. During one stimulus sequence, which consists of four current combinations, one pole in turn acts as a cathode and one pole on the opposite side as an anode. The result is four excitation compartments around the nerve. Combined stimulation of all quadrants of the nerve (each at 5 to 6 Hz) results in activation of the diaphragm near its optimum fusion frequency of 20 to 25 Hz, resulting in smooth contraction of the diaphragm.^{25,52}

By this method, the stimulation frequency of individual axons is less than with monopolar stimulation. The slower stimulus frequency should provide more time for recovery, improve the endurance characteristics of the diaphragm, and shorten the conditioning process, compared to conventional unipolar stimulation.^{25,52,53} This technique had been approved by the FDA through an investigational device exemption. As of October 2005, however, the investigational device exemption study has been terminated. Therefore, this device is no longer available in the United States.

Mediimplant Biotechnisches Labor

The Mediimplant Biotechnisches Labor (Vienna, Austria) system is also differentiated by a unique electrode design.²⁶ A complex microsurgical technique involving placement of four electrode leads around each phrenic nerve is required. The nerve tissue between each electrode lead comprises different stimulating compartments, only one of which is stimulated during any single inspiration. The various compartments are stimulated in sequence during subsequent inspirations (carousel stimulation).⁵⁴ Sixteen different electrode combinations can be adjusted individually for each nerve. Similar to the Atrotech device, only a portion of the nerve is stimulated at any given time, allowing more recovery time and therefore less chance for the development of fatigue compared to the unipolar design. Availability of this system is limited predominantly to Austria and Germany.

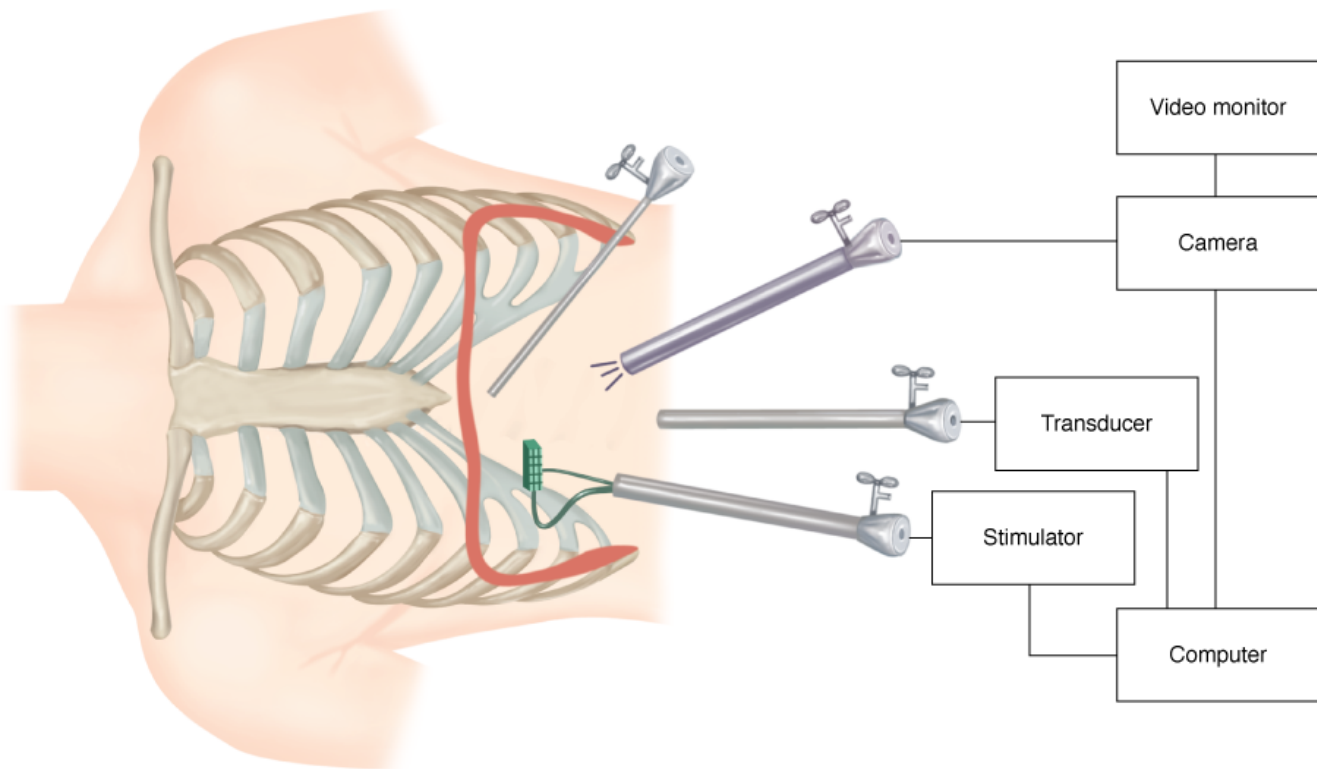
Intramuscular Diaphragmatic Pacing System

The phrenic nerves can also be activated via placement of electrodes directly into the diaphragm (Fig. 62-1B).^{27,28,55,56} The major advantage of this method compared to direct phrenic nerve stimulation systems is that electrode placement does not require a thoracotomy. A thoracotomy is associated with significant perioperative morbidity, requiring an inpatient hospital stay and high cost. These disadvantages have limited the number of patients undergoing this procedure and are significant obstacles for patients undertaking phrenic nerve pacing. Intramuscular diaphragm electrodes can be positioned using minimally invasive laparoscopic techniques. This procedure can be performed on an outpatient basis or with an overnight observational stay, significantly reducing costs.^{27,28,56,57} The risk of nerve injury is virtually eliminated because this procedure does not require manipulation of the phrenic nerve. Postoperative pain is less for patients with CHS and recovery times are reduced significantly.

Synapse Biomedical

Conventional laparoscopy is employed for electrode placement of the Synapse Biomedical (Oberlin, OH) system.⁵⁸ Four laparoscopic ports are required to provide access to the abdominal cavity for visualization, insufflation of the abdominal cavity, diaphragm mapping, and insertion of the implant tool (Fig. 62-2). Specially designed surgical tools and intramuscular electrodes are required for implantation.^{27-29,59-61} Two intramuscular electrodes are implanted into each hemidiaphragm near the phrenic nerve motor points (Fig. 62-3).^{62,63} A mapping procedure is required for appropriate electrode placement.^{27,28} By this method, full-time respiratory support can also be maintained in ventilator-dependent tetraplegic patients (both children and adults) with success rates similar to those of direct phrenic nerve stimulation.^{19,27,28,34,64} It is important to note that despite the fact that electrodes are positioned within the diaphragm muscle, the mechanism of diaphragmatic activation by this method is stimulation of the phrenic nerve motor points. This technique, therefore, represents a form of phrenic nerve stimulation, rather than direct diaphragm muscle stimulation.

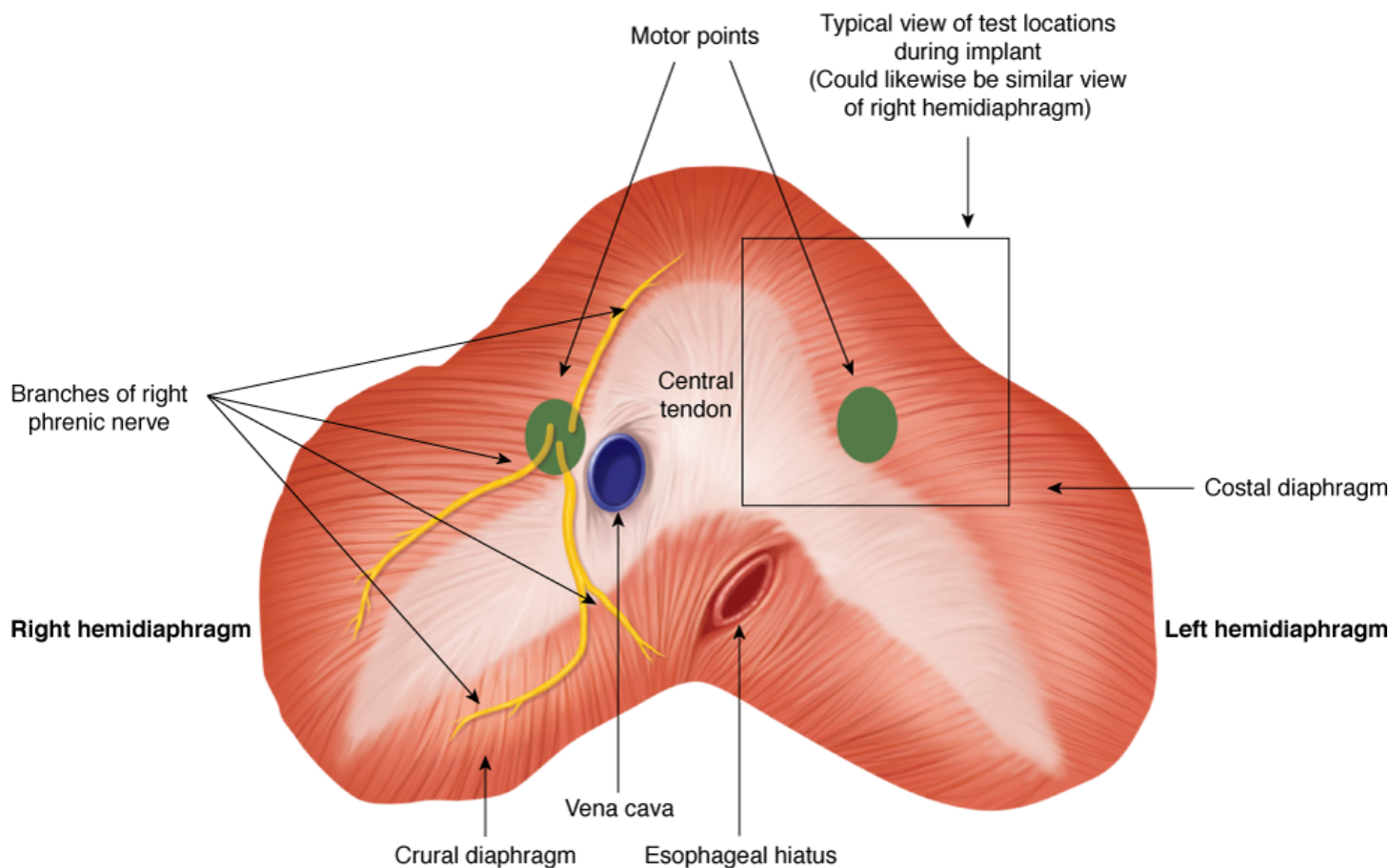
Figure 62-2



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com
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Schematic of laparoscopic implant materials required for implantation of intramuscular diaphragmatic electrodes. Four laparoscopic ports are necessary to provide access to the abdominal cavity. Ports are necessary for visualization, insufflation of the abdominal cavity, diaphragmatic mapping, and insertion of the electrode implant tool. (Reprinted with permission of the American Thoracic Society. Copyright © 2012 American Thoracic Society. DiMarco AF, Onders RP, Kowalski KE, et al. Phrenic nerve pacing in a tetraplegic patient via intramuscular diaphragm electrodes. *Am J Respir Crit Care Med*. 2002;166:1604–1606. Official Journal of the American Thoracic Society.)

Figure 62-3



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Schematic of the anatomy of the diaphragm from the abdominal surface and entrance points of each phrenic nerve into each hemidiaphragm (phrenic nerve motor points). (Reprinted with permission of the American Thoracic Society. Copyright © 2012 American Thoracic Society. DiMarco AF, Onders RP, Kowalski KE, et al. Phrenic nerve pacing in a tetraplegic patient via intramuscular diaphragm electrodes. *Am J Respir Crit Care Med*. 2002;166:1604–1606. Official Journal of the American Thoracic Society.)

Combined Intercostal and Diaphragmatic Pacing

Many patients with cervical spinal cord injury have damage to one or both phrenic motor neuron pools in the spinal cord and/or phrenic rootlets, and therefore cannot be offered DP.^{31,65,66} By placing electrodes epidurally on the ventral surface of the upper thoracic spinal cord, however, the inspiratory intercostal muscles of the upper rib cage can also be stimulated to produce large inspired volumes.^{67–69} Moreover, gas exchange during intercostal breathing alone is comparable to diaphragmatic breathing.⁷⁰

In initial clinical trials, in patients with absent diaphragmatic function, stimulation of the intercostal muscles alone produced inspired volumes of similar magnitude to those resulting from activation of a single hemidiaphragm.⁶⁹ Inspired volume, however, was not sufficient to support ventilation for prolonged periods. In subsequent trials in tetraplegic patients with unilateral diaphragm function, however, stimulation of the intercostal muscles in combination with unilateral phrenic nerve stimulation was successful in providing long-term ventilatory support.⁷¹

Side effects of intercostal muscle stimulation included mild flexion of both hands and contraction of the muscles of the upper torso, which was well tolerated.^{69,71} Intercostal pacing may be a useful adjunct to enhance tidal volume in patients whose inspired volume with phrenic nerve pacing is suboptimal.

This technique is not yet commercially available but has received approval by the FDA through an investigational device exemption.

Patient Selection

As mentioned, DP provides clinical benefit in two patient groups: ventilator-dependent tetraplegics^{8,11,34} and patients with CHS.^{5,6,72} Although phrenic nerve pacing has been tried in patients with chronic obstructive pulmonary disease to prevent ventilatory depression associated with oxygen

administration,⁷³ noninvasive ventilator support is of equal or greater effectiveness.

In all patients, significant lung, chest wall, cardiac, or primary skeletal muscle disease must be excluded because these conditions may preclude successful pacing.^{40,47,74}

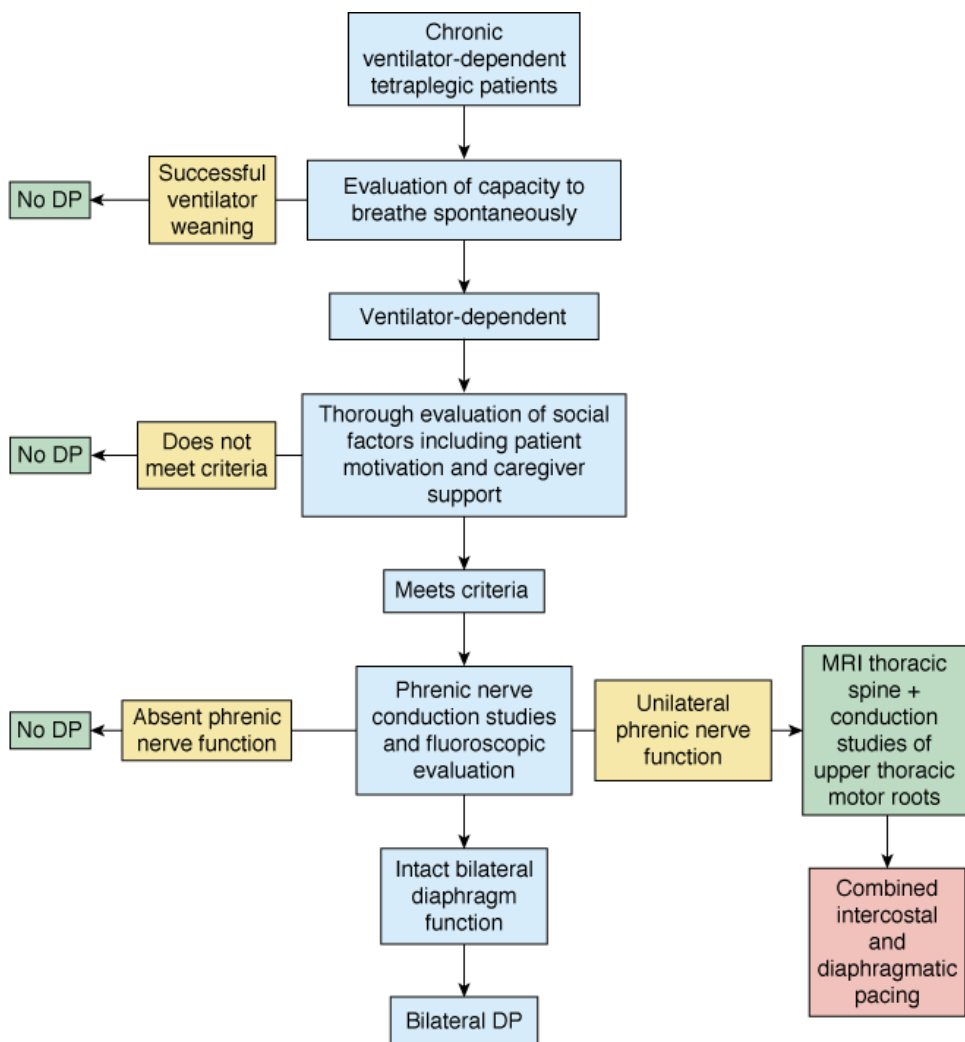
Ventilator-Dependent Tetraplegia

Respiratory failure is common following acute cervical spinal cord injury.^{65,66} Fortunately, most patients achieve significant recovery following their initial presentation and are able to breathe spontaneously. Nonetheless, approximately 4% require lifelong respiratory support.⁵⁷ Although the time course of recovery is variable, the degree of impairment is likely permanent 12 to 15 months following injury.^{66,75} Given the high cost and the invasive procedure required, direct phrenic nerve DP should not be considered before this time. However, given the minimally invasive nature of intramuscular DP, and the significant advantages of DP compared to mechanical ventilation, much earlier placement of this system should be considered. In the event that the clinical status of the patient improves to the point that spontaneous breathing is restored, the DP system can be removed.

Vigorous attempts should be made to wean tetraplegic patients from mechanical ventilation. Vital capacity measurements of less than 10 mL/kg suggest inadequate inspiratory muscle function to maintain spontaneous breathing.⁶⁶ In some patients with sufficient inspiratory muscle strength, however, noninvasive ventilator support may be a suitable alternative to DP.^{76,77}

Individual psychosocial conditions are important factors in the ultimate success of phrenic nerve pacing.^{27,31,34,49} A high level of motivation and cooperation of the patient and family members is imperative. A home situation in which the family unit is anxious to improve the mobility, social interaction, occupational potential, and ability of the patient to function independently is optimal. Consequently, a psychosocial evaluation should be conducted before any technical evaluation. Figure 62-4 shows the evaluation process of potential candidates for DP.

Figure 62-4



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Evaluation of potential candidates with cervical spinal cord injury for diaphragmatic pacing.

Central Hypoventilation Syndromes

Patients with CHS consist mainly of infants and children. Pacing is not instituted before 6 to 12 months of age to allow time to identify other potential abnormalities and determine the full extent of the ventilatory deficit.⁴ CHS can be congenital¹⁻³ or secondary to brainstem injury from injury, bleeding, tumor, encephalitis, or Arnold-Chiari malformations.^{4,11} To assess the presence and degree of hypoventilation, arterial blood gases, nocturnal polysomnograms, and ventilatory responses to hypercarbia and hypoxia should be performed.

The diagnosis of congenital CHS is usually made shortly after birth. Many of these patients have normal wake ventilation and therefore require DP only during sleep.³⁻⁵ Patients who require full-time ventilator support are paced during the day and maintained on mechanical ventilation during sleep. A psychosocial evaluation of these patients should also be performed.

Evaluation of Phrenic Nerve Function

Before instituting DP, a thorough evaluation of phrenic nerve function is mandatory in all patients.^{10,23,27,72} Phrenic nerve function can be assessed by measurements of nerve conduction times.^{45,78,79} Diaphragmatic action potentials are monitored by surface electrodes positioned between the seventh and ninth intercostal spaces. The phrenic nerves can be electrically stimulated by surface electrodes or monopolar needle electrodes at the posterior border of the sternocleidomastoid muscle at the level of the cricoid cartilage. Electrical current is applied with single pulses of gradually increasing intensity until a supramaximal M wave is seen. With stimulation, the abdominal wall expands outward. Phrenic nerve conduction time can be determined by measuring the time interval between the applied stimulus and onset of the compound muscle action potential (CMAP). Phrenic nerve function can also be assessed by cervical magnetic stimulation of the phrenic nerves.^{80,81} Mean onset latency is 7 to 9 milliseconds in normal adults.^{45,78,79} Mild prolongation of conduction times (up to 14 milliseconds), however, does not preclude successful pacing.¹² In children, latencies

are significantly shorter at 2.2 milliseconds at 6 months of age, but increase to 4.2 milliseconds between 5 and 11 years of age.^{7,82} The magnitude of the CMAP is a less reliable indicator of phrenic nerve function compared to conduction time.^{32,78}

Fluoroscopy is also a useful method to assess phrenic nerve function. The diaphragm should descend at least 3 to 4 cm following stimulation.^{10,31,45} In children with CHS, diaphragmatic descent of at least two rib spaces following spontaneous effort indicates adequate phrenic nerve function.⁴ In this population, phrenic nerve stimulation is reserved for patients in whom the fluoroscopic results are questionable.

A recent study⁸³ evaluated the use of diaphragmatic compound muscle action potentials (CMAPs) to evaluate phrenic nerve function. The investigators found that diaphragmatic CMAPs were recorded only when the diaphragm was observed to move on fluoroscopic examination. These results suggest that assessment of the presence of diaphragmatic CMAPs can be used interchangeably with fluoroscopic evaluation to assess phrenic nerve integrity. Because of false-positives and false-negatives associated with conduction time measurements, and technical challenges associated with this test, fluoroscopy should be performed on all patients being evaluated for DP (personal observation).

Phrenic nerve function can also be assessed by measurements of transdiaphragmatic pressure (Pdi) (i.e., the pressure difference across the diaphragm).⁴³ By this method, small balloon-tipped catheters are placed into the esophagus and stomach to determine intrathoracic and intraabdominal pressures, respectively. Single-shock stimulation results in Pdi values of approximately 10 cm H₂O for each side in normal subjects.⁸⁴ Because of diaphragmatic atrophy, this value may be reduced significantly in ventilator-dependent patients. In spontaneously breathing individuals, diaphragmatic strength can be assessed by measuring Pdi during a maximal sniff maneuver.⁸⁴

Implementation

The various surgical techniques to implant electrodes necessary for DP have been described elsewhere^{9,12,85} and are beyond the scope of this chapter.

DP should not commence until 10 to 14 days postoperatively to allow resolution of inflammation and edema at the electrode–nerve interface and initial healing of surgical wounds.¹⁰ In patients who have been maintained on mechanical ventilation for prolonged periods, the transition to DP requires a gradual conditioning period, because of diaphragmatic atrophy.^{11,23,24,86} Too rapid institution of pacing carries the risk of diaphragmatic fatigue and possible injury.

Determination of Stimulus-Output Values

Ideally, several parameters should be determined initially and then monitored regularly.^{12,32} These include stimulus threshold values (minimum stimulus amplitude that results in visible or palpable diaphragmatic contraction) and supramaximal amplitudes and frequencies (lowest stimulus parameters that result in maximum inspired volume production). Changes in airway pressure during airway occlusion also provide a useful assessment of diaphragmatic force. The magnitude of inspired volumes and force generation gradually increase during the conditioning phase. Plateaus in these indices over time suggest that the conditioning phase is complete and optimum diaphragmatic function has been achieved. Assessment of diaphragmatic action potentials may also be useful in the determination of maximum stimulus amplitude.⁴

Determination of Stimulus Parameters to Achieve Adequate Ventilation

Utilizing supramaximal amplitudes, tidal volume is adjusted by changing stimulus frequency. With respiratory rates (stimulus train rates) between 8 and 14 breaths/min, stimulus frequency is adjusted to achieve inspired volumes resulting in partial pressure of carbon dioxide (P_{CO_2}) values in the low normal range. Stimulus frequency should be set at the lowest level possible and should not exceed 20 Hz. Respiratory rate and tidal volume are further adjusted for patient comfort. Inspired volume measurements should be made both in the supine and upright postures. While sitting, spinal cord-injured patients should wear a snug fitting abdominal binder because of the shorter diaphragm length and higher resting lung volume associated with this posture. Higher stimulus parameters may be required in this posture to achieve adequate ventilation.

Institution of Diaphragmatic Pacing

The methods of transition from mechanical ventilation to pacing are somewhat arbitrary. With the objective of achieving full-time DP as quickly as possible, but without the development of diaphragmatic fatigue, the following protocol is suggested. Utilizing the above-mentioned stimulus parameters, continuous bilateral DP is initially provided until significant blood gas derangements (reductions in oxygen saturation or elevations in end-tidal P_{CO_2}), reductions in inspired volume generation (monitored every 5 to 10 minutes), or patient discomfort related to air hunger are observed. Pacing is then initiated for a somewhat shorter time period (approximately 5 minutes) every hour during the day, for the first week. This assessment is

repeated weekly and a new pacing schedule is applied accordingly. After full-time pacing is achieved during wakefulness, pacing is provided during sleep. Most patients cap their cuffless tracheostomy tube while pacing during the daytime. During sleep, however, airflow through the tracheostomy should be maintained to prevent upper airway obstruction. This occurs secondary to the dyssynchrony between upper airway and diaphragmatic contraction during sleep.^{5,23}

Previous studies have demonstrated that chronic diaphragmatic stimulation at high frequencies can be associated with myopathic changes and consequent reductions in force generation.^{8,34,87} Consequently, it is important that DP occur at low stimulus frequencies (<20 Hz), which convert the diaphragm from a mixed-fiber-type population to predominantly high-oxidative, slow-twitch fatigue-resistant type I fibers.^{8,88} As the conditioning phase progresses, stimulation frequencies and respiratory rates should be gradually reduced to the lowest values that maintain adequate ventilation and patient comfort. With the Avery system, stimulation frequency can usually be reduced to 7 to 9 Hz with respiratory rates of 6 to 12 breaths/min in the supine position.⁸ Less adjustment is usually required with the Atrotech device secondary to the initial application of low frequencies via sequential nerve stimulation. With intramuscular stimulation, stimulation is initiated with four electrodes at 20 Hz. Stimulation frequency can be reduced to as low as 11 Hz with stimulation of only two electrodes.²⁷ Infants and young children require higher respiratory rates in the range of 20 breaths/min, but can be maintained with lower inspiratory times of 0.6 to 0.9 second compared to the requirement of 1 to 1.4 second in adults.^{5,89} Conditioning time can vary between 4 weeks and several months. In most instances, however, conditioning is accomplished within 6 to 10 weeks.

In patients with CHS who employ DP only during sleep, DP should be adjusted in a sleep laboratory.⁴ Given the high compliance of the rib cage and higher ventilatory requirements, bilateral pacing is always necessary to support ventilation in infants and young children.^{4,5} In patients undergoing daytime pacing, pacing parameters must be reevaluated because ventilatory requirements are typically higher during wakefulness.

Combining Pacing with Mechanical Ventilation

Full-time pacing is generally not advisable in infants and small children to avoid fatigue and the risk of permanent injury to the diaphragm and phrenic nerves.^{3-5,89} Consequently, children with tetraplegia and those with CHS who require full-time ventilator support utilize DP during the day and are maintained on mechanical ventilation at night.³ Likewise, adult patients who do not achieve inspired volumes sufficient to maintain full-time pacing can be maintained on ventilator support at night. Because the major advantages of DP are realized during the day and most patients still require a tracheostomy, part-time DP does not detract significantly from the benefits of this device.

In some instances, patients cannot tolerate significant time off mechanical ventilation and initial inspired volumes achieved by pacing are insufficient to maintain adequate ventilation for even short periods.⁵⁹ In these patients, the initial phase of muscle reconditioning can be performed in conjunction with mechanical ventilation. This can be accomplished by setting the ventilator in the assist mode. The negative inspiratory pressure generated at the tracheal opening by pacing can be used to trigger the ventilator.

Effect of Diaphragmatic Pacing on Patient Outcome

Clinical studies reveal that DP is a feasible and effective means of providing ventilatory support in patients with tetraplegia and CHS. In most patients requiring full-time ventilatory support, DP can be utilized as the sole mode of respiration.^{8,11,27,34}

The actual success rate of DP, however, is difficult to ascertain because most patients have undergone this technique at centers where patient numbers are small.^{35,41,90} Moreover, previous large series have evaluated the outcome of DP while technology was still in development.^{23,91-93} Consequently, these results are not applicable to systems available today.

In a recent report, the clinical outcome of patients whose DP systems were implanted before 1981 was compared to patients implanted between 1981 and 1987.³⁴ The earlier group underwent high-frequency DP; the success rate in terms of achieving full-time pacing was small because of the development of diaphragmatic fatigue. The latter group underwent low-frequency stimulation at low respiratory rates; DP was successful in each of the twelve patients, stressing the importance of current pacing regimes. Six patients continued pacing for a mean duration of 14.8 years. The other six patients did not achieve sustained long-term pacing because of lack of adequate financial and social support or concomitant medical conditions. The results of this analysis indicate that the long-term success of DP depends heavily on adherence to strict criteria for patient selection. High success rates have also been observed in infants and children in whom a more modest goal of part-time pacing (<15 hours/day) was achieved.⁴

The more recently developed quadripolar electrode design and receiver (Atrotech OY) is associated with high success rates. Based on analysis of sixty-four patients, successful pacing was achieved in 94% of pediatric and 86% of adult patients.⁵³ At the time of study, however, the average duration of pacing was approximately 2 years.

Complications and Side Effects

Technical developments and clinical experience with DP over the past 20 years have markedly reduced the incidence of complications and side effects. With modern-day equipment, appropriate patient selection, proper use of stimulus paradigms, and adequate patient monitoring, complications are few. As with any life-support system, patients must be carefully monitored because pacemaker failure can have catastrophic consequences. For this reason, all patients who are unable to maintain adequate spontaneous ventilation for prolonged periods should have emergency access to mechanical ventilation.

Several technical problems can cause reductions in volume production (Table 62-3). The most common cause of mechanical failure is loss of battery power. With current pacing systems, however, routine maintenance requires regular battery changes and recharging schedules. Low-battery alarms are also present. Breakage of the external antenna wires at stress points, either near the connection to the transmitter or more commonly at the connection to the antennae, can occur. With the intramuscular DP system, wire breakage can occur at the skin exit site.

Table 62-3: Complications and Side Effects of Diaphragmatic Pacing

A. Technical malfunction
1. External components
a. Battery failure
b. Breakage of antenna wires
2. Implanted components
a. Receiver failure
b. Electrode malfunction
c. Breakage of implanted connecting wires
B. Infection
1. Receiver site
2. Electrode site
C. Mechanical injury to the phrenic nerve
1. Iatrogenic injury at the time of surgery
2. Late injury as a result of scar formation and/or tension on the nerve
D. Upper airway obstruction after tracheostomy closure
E. Paradoxical movement of the upper rib cage, particularly in children

Concerning the internally implanted components with direct phrenic nerve stimulation systems, failure of the radiofrequency receiver was a fairly common problem with older systems, secondary to leakage of body fluid through the epoxy encapsulation. Wire breakage within the receiver was also an issue.⁹¹ With older systems, this problem often occurred within 5 years of implantation.^{23,47,91} With improvement in housing materials, receiver life has been extended significantly and reports of receiver failure are much less common.⁵³ Electrode wire malfunction or breakage is also much less common but can occur unpredictably at variable time periods following implantation. The most recent analysis of the quadripolar electrode system in patients who had undergone DP for approximately 2 years, demonstrated an electrode failure rate of 3.1%.⁵³ Failure rates were similar in tetraplegics and patients with CHS, and in adults and children. Failure of one or more of the four electrode combinations was fairly common, but usually did not interfere with successful pacing.⁵³

Iatrogenic injury of the phrenic nerve can occur secondary to mechanical trauma during electrode implantation or secondary to subsequent tissue reaction and fibrosis around the electrode.^{11,23,88} When performed at centers, however, by surgeons with expertise in DP and use of monopolar or quadripolar electrodes, rather than the older bipolar cuff electrodes that encircled the nerve, phrenic injury is infrequent.⁵³

As with any foreign body, implantation of the internal components carries some risk of infection. Since the institution of this technique in the 1970s, infection rates have been relatively constant in the range of 3% to 5%.^{22,53,91} The higher infection rate observed in active children with CHS is thought to occur as a result of the greater likelihood of local trauma in this population.⁵³ The development of infection usually requires the removal of all implanted materials.

Although there are some reports of successful tracheostomy closure following institution of DP, these are very uncommon. During the daytime, the state of wakefulness is associated with synchronous activation of the upper airway muscles and diaphragm during DP. During sleep, however, there is reduced activation of the upper airway dilator muscles and a greater tendency toward asynchronous upper airway muscle contraction resulting in

upper airway obstruction, a form of obstructive sleep apnea.^{5,94,95} With few exceptions, therefore, patients undergoing DP require a patent tracheostomy for nocturnal use. Maintenance of a tracheostomy also facilitates application of mechanical ventilation in the event of pacemaker malfunction or instances of increased respiratory demand such as acute infections.

With direct phrenic nerve pacing systems, monopolar stimulation may interfere with demand-type cardiac pacemakers. For this reason, if a cardiac pacemaker is present, it is advisable to use bipolar electrodes for DP.⁹⁶ The electrodes should be at least 10 cm from the cardiac pacemaker.^{96,97} The presence of cardiac pacemakers, however, is not a contraindication for use of the intramuscular DP system. In twenty tetraplegic subjects with cardiac pacemakers and DP, there were no device-to-device interactions.⁹⁸

In young children, paradoxical movement of the rib cage may occur as a result of the high chest wall compliance, resulting in reduced inspired volume generation. Consequently, bilateral pacing is required because of the paradoxical movement of the contralateral diaphragm and chest wall during DP.^{4,23}

Finally, strong magnetic fields, such as those that occur with magnetic resonance imaging scanning, can override the electronic circuitry of pacing systems. The substantial energy transmission associated with this scanning could be transmitted to the electrode resulting in phrenic nerve injury. The pacing system's internal components would also be attracted to the magnet. Exposure to electrotherapeutic devices that generate strong radiofrequency fields should also be avoided because they may interfere with the pacing system.

Monitoring

Unlike mechanical ventilators, DP systems have no alarm systems that reflect inadequate levels of ventilation. When not receiving direct attendance by caregivers, patients should employ a pulse oximeter as a monitor for possible pacemaker malfunction or inadequate tidal volume generation secondary to patient-related issues.

Bedside Evaluation

For all patients supported by DP, breathing should be comfortable and effortless. Caregivers should evaluate pacemaker function on a routine basis and also in situations that give rise to dyspnea. On palpation of the chest wall, inspiration should be associated with vigorous lateral expansion of the lower rib cage and anterior abdominal wall, bilaterally. Attachment of a spirometer to the tracheostomy tube allows easy measurement of tidal volume. The transmitter enables separate stimulation of each hemidiaphragm, allowing evaluation of each side independently. Adequacy of ventilation should be assessed with pulse oximetry and end-tidal CO₂ measurements.^{31,32,43,70}

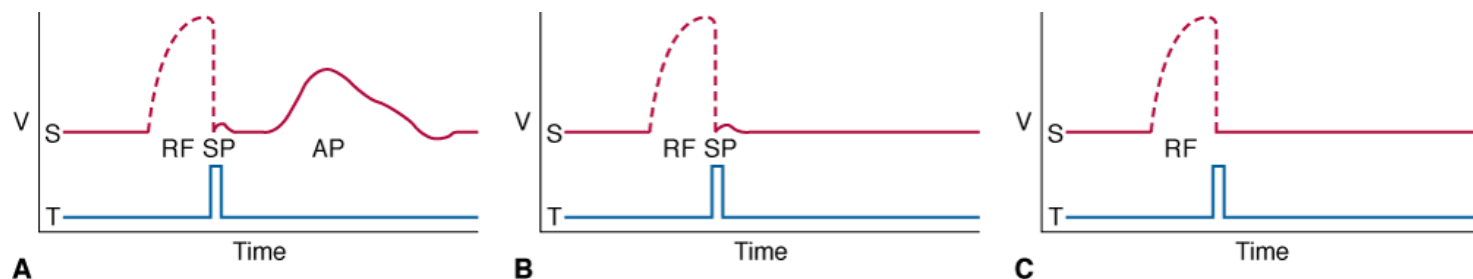
Troubleshooting

If inspired volume is inadequate, the function of the external components should be evaluated first. Initially, the batteries should be replaced because this is the most common cause of pacemaker malfunction. Subsequently, the antenna should be replaced. If function is not restored, the back-up transmitter should then be changed.⁴

The occurrence of abnormal lung mechanics can also cause reductions in inspired-volume generation during DP. For example, most tetraplegics have a markedly reduced ability to cough and consequently accumulate airway secretions that require regular evacuation. Retained secretions cause increases in airway resistance and the development of atelectasis with secondary reductions in lung compliance. These mechanical derangements will reduce inspired-volume generation. Removal of secretions usually results in prompt improvement. Likewise, bronchitis and pneumonia will also result in reductions in inspired volume. If reductions in inspired volume are persistent, therefore, chest radiography may be necessary. This test is also useful to evaluate for wire breakage and electrode position.

With the direct phrenic nerve stimulation systems, evaluation of the internal components can be performed according to previously described techniques.⁹¹ Surface electrodes are placed at the costal margin to record the pacemaker stimulus pulse and diaphragmatic action potential. The signals are amplified and recorded on an oscilloscope. Figure 62-5 is a schematic of the signals obtained from the chest leads. The DP system is functioning properly if the radiofrequency signal from the transmitting antenna, stimulus pulse from the phrenic nerve electrode, and diaphragmatic action potential are seen on the oscilloscope. If the radiofrequency signal and stimulus pulse are observed without the action potential, this indicates that the wire insulation is no longer intact, the phrenic nerve is not in contact with the electrode, or the phrenic nerve has been damaged. If the stimulus pulse and action potential are both not seen, the receiver is not functional.

Figure 62-5



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com
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A. Schematic of an oscilloscopic tracing showing a properly functioning diaphragmatic pacing system. The lower tracing *T* represents the output from an extra receiver that was used to trigger the oscilloscope sweep. The upper trace (*S*) represents the signal obtained from the percutaneous leads on the chest. Shown in sequence are the radiofrequency signal (*RF*) from the transmitting antenna, the stimulus pulse (*SP*) from the electrode on the phrenic nerve, and the compound action potential (*AP*) from the diaphragm. **B.** Schematic of a malfunctioning pacing system. The tracing is the same except for the absence of the action potential. Breakage of wire insulation, lack of adequate phrenic nerve/electrode contact, or phrenic nerve damage are consistent with this finding. **C.** Schematic of another type of malfunctioning pacing system. In this case, the action potential and radiofrequency signal are absent. This finding suggests receiver malfunction. (Used, with permission, from Weese-Mayer et al.⁹⁴)

Important Unknowns

The impact of DP, compared to mechanical ventilation, on long-term survival is unknown. Carter et al,⁹⁹ however, compared survival rates in a retrospective analysis. Overall survival rates were similar between groups, but the patients on mechanical ventilation expired earlier than did patients maintained on DP. In a more recent clinical trial of sixty-four patients over a 20-year period, there was no significant difference in longevity between use of DP and mechanical ventilation.³⁹ Respiratory tract infections, however, were a more common cause of death in the mechanical ventilation group.

In the absence of depressed mental status and weakness of oropharyngeal muscles, some investigators have demonstrated that direct airway pressure methods, including mouth and nasal intermittent positive-pressure ventilation, can maintain adequate ventilation in tetraplegic patients.^{74,75} These modalities offer the significant advantage of tracheostomy closure. Given the high cost and invasive nature of phrenic nerve implantation, these modalities may be better suited for some individuals. Newer methods of electrode placement, however, may swing the advantage in favor of DP.^{27,28}

The Future

Although existing DP systems provide substantial lifestyle advantages to patients compared to mechanical ventilation, current systems are somewhat cumbersome and have significant disadvantages and limitations. With the ultimate goal of complete restoration of normal respiratory system function, several refinements are needed.

Perhaps most significant is the future development of systems that synchronize upper-airway muscle and diaphragmatic contraction, which would eliminate upper-airway obstruction during DP.^{94,95} One option is to use an upper-airway muscle signal to trigger diaphragm activation. Such a device would not only eliminate the need for a tracheostomy but also provide ventilation on demand, allowing for changes in ventilatory requirements. A totally implantable system has not yet been developed, in part because of the high energy requirements of phrenic nerve pacing. A totally implantable system, similar to cardiac pacemakers, would eliminate the need for attachment of materials to the body surface, connection to a transmitter box, and battery changes. This development would further improve patient convenience and mobility.

DP in tetraplegic patients often requires a long reconditioning program because of disuse atrophy. Implantation of less-invasive intramuscular electrodes soon after injury would allow more tetraplegic patients to use DP in place of mechanical ventilation. In patients in whom weaning is possible, electrodes can subsequently be removed without injury. Using this technique, it is also conceivable that intensive care unit patients who require prolonged mechanical ventilation with the expectation of eventual recovery may benefit from DP to prevent diaphragmatic atrophy and consequent difficulty in weaning.

Finally, in patients with bilateral phrenic nerve injury, intercostal to phrenic nerve transfer may restore phrenic nerve viability and provide patients with the option of DP.¹⁰⁰

Summary and Conclusions

In patients with ventilator-dependent tetraplegia and CHS, DP represents a practical method of ventilatory support with significant health and lifestyle advantages compared to mechanical ventilation. Advantages include increased mobility and level of independence, improved speech, improved sense of smell, and reduced anxiety and embarrassment associated with mechanical ventilation. The DP systems are smaller and more portable than ventilators and require less maintenance. Patients, however, must be carefully screened for social factors, including level of patient motivation and caregiver support, as well as coexisting medical conditions. Adequacy of phrenic nerve function must also be thoroughly evaluated preoperatively. In some patients, other means of support such as noninvasive ventilation may be more appropriate. Clinical studies examining methods of activating the inspiratory intercostal muscles are underway. These new techniques are likely to increase the number of eligible patients and expand the clinical indications for DP.

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Disclosures

Dr. DiMarco is a founder and owner of Synapse Biomedical, LLC, a manufacturer of DP systems.

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